Data Science Project

HOW AI IS RESHAPING DEVELOPER LANGUAGE CHOICES OVER TIME

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TECHNICAL OVERVIEW

Data collected from GitHub REST API (2020–2025).

~5750 repositories collected

Collected features: stars, forks, languages, topics, contributors, timestamps, etc.

Steps

- **Data cleaning**
- **Temporal segmentation (pre/post ChatGPT)**
- **Feature engineering**
- **Clustering, regression, hypothesis testing**
- **Evaluation and interpretation**

DATA PREPARATION

Temporal Segmentation: period = before/after Nov 30, 2022

Feature Selection: stars, forks, contributors_count, created_year, primary_language

Regression Target:

- Derived secondary_bytes (largest non-Python language bytes)
- Log-transform using log1p to reduce skew

Encoding: LabelEncoder for categorical features

Splits: 80% train / 20% test

ALGORITHMS AND MODELLING TECHNIQUES USED

1. Clustering
- K-Means → good for spherical groups, used to detect natural structure
- DBSCAN → density-based, tested for non-spherical clusters
2. Regression
- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor (non-linear, robust, high-performance)
3. Statistical Hypothesis Testing
- One-sample and two-sample t-tests ($\alpha = 0.05$)
- Used to compare pre- vs. post-ChatGPT language shifts

CLUSTERING RESULTS

K-Means

- Best K=4 (Elbow Method)
- Silhouette Score = 0.383 (moderate separation)
- Identified outliers: high-star / high-fork repositories

DBSCAN

- Found one dense core cluster
- Only 1.10% classified as noise
- Confirms AI repos form one large connected cluster

REGRESSION RESULTS

Model	MSE	R^2
Baseline	**23.82**	**-0.000**
Linear Regression	**15.78**	**0.337**
Decision Tree Regressor	**3.70**	**0.845**
Random Forest Regressor	**2.11**	**0.912**

Key points:

- Random Forest = best-performing model
- Explains 91.2% of variance
- Shows strong non-linear relationships
- Secondary language size is predictable based on stars/forks/contributors

HYPOTHESIS TESTING FINDINGS

Python Creation Decline

- Before ChatGPT: 47.86%
- After ChatGPT: 40.41%
- A significant drop → AI ecosystem becomes more diverse
- (More JS/TS projects)

Python Popularity increase

- Average stars: +168% increase
- Mean stars 2171 → 5812
- $p < 0.001$

Interpretation

- Python shifts from "most common" → high-impact / elite language
- LLMs attract non-Python developers into AI projects

EVALUATION METRICS USED

Clustering

- Silhouette Score
- Noise % for DBSCAN

Regression

- Mean Squared Error (MSE)
- R^2 Score

Hypothesis Tests

- p-values
- Effect size and percent change
- Statistical power analysis

TECHNICAL HURDLES ENCOUNTERED

Data Collection Issues

- API rate limits → solved using personal access token
- Long extraction time → proposed storing periodic snapshots

Modelling Challenges

- Highly skewed data → log transform fixed regression skew
- Very imbalanced languages → classification abandoned
- Clustering overlap → DBSCAN showed clustering not meaningful for most repos

IMPROVEMENTS & FUTURE WORK

Data Improvements

- Handle imbalance with SMOTE or class weighting
- Expand dataset beyond top-starred repos
- Collect commit history, README text, code complexity metrics

...

Feature Improvements

- NLP on description/topics → improves predictive signal
- Use embeddings instead of manual categories

Model Improvements

- Try XGBoost or LightGBM
- Better cluster validation (e.g., hierarchical clustering)

CONCLUSION

The AI ecosystem is expanding toward JS/TS-based tools, enabled by LLMs

Random Forest regression reveals strong relationship between repository success and secondary language complexity

Statistical evidence is strong (large N, high power)

ChatGPT shifted the developer ecosystem:

- Python proportion ↓, Python impact ↑

THANK YOU